Question 3

a)

i) firstly, the frequency range of sound that humans can hear is 20-20 KHZ. According to Nyquist theorem, the sampling rate should be at least two times larger than the maximum frequency in sounds. Therefore, for good quality sound the sampling rate must be at least 40 KHz.

Secondly, we know that the higher sampling rate, the less possibility of aliasing and overlapping, because in reality we can't produce the ideal filter and should consider the transition bandwidth. Therefore, we choose the sampling rate of 44.1 KHz larger than 40 KHz.

ii)

iii) for voiced sounds, their spectrum shows characteristic maxima, and these maxima is formants, implying that energy in the region is relatively concentrated. Because different people have different positions and ways of pronunciation, the timbre of the voice is then different, and this difference can be directly expressed through the formant. Therefore, by comparing the formant collected in the database with the current formant, speech recognition can be carried out.

In spectrogram, formants can be expressed by the black bar parallel to the horizontal timeline.

b)

i) throw away some of the bits used to represent color information, but these bits are out of human eye's perception.

ii) achieve high compression. Because quantization means subtracting DCT coefficients by the quantization table and rounding off, which means if the table contains high values, the DCT coefficients, especially those for high-frequency components that are typically small, will easily round down to 0. When those value achieve 0, they will be thrown away and don't need to be stored, getting a high compression in total.

c)

i) because for the top left macroblock, there is no matching macroblock to be found in successive frames, so there is no prediction can be made. We will use intra-frame encoding.

ii) because for the bottom right macroblock, there is one similar macroblock to be found in the future frame (the right frame), so we can make prediction through this reference frame. And we will use inter-frame encoding to encode changes, that is the different values between the two macroblocks (error terms) and a motion vector.